

Binary Outcome Models: Endogeneity and Panel Data

ECMT 676 (Econometric II) Lecture Notes

TAMU

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Topics

Issues in binary response models:

- Endogeneity
 - ▶ Continuous endogenous x
 - ★ Control function approach
 - ★ IV probit approach
 - ▶ Binary endogenous x
- Panel data
 - ▶ RE
 - ▶ CRE
 - ▶ FE

Binary response model with endogeneity

- Endogeneity arises naturally in the *latent variable model*
- LPM (linear probability model) with 2SLS is a handy solution for binary response with endogeneity. But it does not allow for individual-specific marginal effects
- We will deal with probit/logit with endogeneity
- Why endogeneity causes problem in binary model? not as transparent as in the linear model

- The binary outcome model:

$$\begin{aligned} y_i &= I\{y_i^* > 0\} \\ y_i^* &= x_i'\beta + u_i \end{aligned}$$

- Now we allow $E x_i u_i \neq 0$ by assuming $u_i | x_i \sim N(q(x_i), 1)$.
- Then

$$\begin{aligned} P(y_i = 1 | x_i) &= P(u_i < x_i'\beta | x_i) \\ &= P(u_i - q(x_i) < x_i'\beta - q(x_i) | x_i) = \Phi(x_i'\beta - q(x_i)) \end{aligned}$$

- Probit of y_i on x_i is not consistent.
- We will see general endogeneity ($\sigma_{uv} \neq 0$) introduces not only *non-constant mean* but also *non-unity variance*.

When the endogenous variable is continuous

A control function (CF) approach

- The part is based on Blundell and Powell (2004 REStud) and also Wooldridge (15.7.2).
- Unanswered question: why not traditional 2 stages?
- The model

$$\begin{aligned}y_{1i} &= I\{y_{1i}^* > 0\} \\ y_{1i}^* &= x_i' \beta_0 + u_i = z_{1i}' \beta + \alpha y_{2i} + u_i\end{aligned}$$

- Now suppose part of x_i (called y_{2i} , assuming as a scalar) is endogenous: $x_i = (z_{1i}', y_{2i})'$.
- The vector of IVs $z_i = (z_{1i}', z_{2i}')'$.
- The reduced form (RF)

$$y_{2i} = z_i' \delta + v_i = z_{1i}' \delta_1 + z_{2i}' \delta_2 + v_i$$

- The parameters (α, β) index the average structural function (ASF) (or, the response probability).
- Again, this is defined by fixing all observed explanatory variables and integrating out the unobservable, u :

$$\begin{aligned} ASF(y_2, z_1) &= E_u\{1[z_1'\beta + \alpha y_2 + u > 0]\} \\ &= \Phi(z_1'\beta + \alpha y_2) \end{aligned}$$

- Thus, α and β are the parameters that appear in the APEs (derivatives of ASF).

- Assume

$$\begin{pmatrix} u_i \\ v_i \end{pmatrix} | z \sim N \left(0, \begin{pmatrix} 1 & \sigma_{uv} \\ \sigma_{uv} & \sigma_v^2 \end{pmatrix} \right)$$

- Endogeneity comes from $\sigma_{uv} \neq 0$.
- Normality of v is not realistic if y_{2i} is binary. So we are assuming for now the endogenous variable is *continuous*.
- Probit has some advantage here over logit, because we can then assume joint normality

- Rivers and Vuong (1998) proposed a "control function (CF)" approach: "which introduces *residuals from the reduced form* for the regressors as covariates in the binary response model to *account for endogeneity*" (BP, 2004).
- Linear projection

$$u = \theta v + e$$

where $\theta = \sigma_{uv} / \sigma_v^2$ and e is also normal.

- Having the linear projection, we get

$$\begin{aligned} I(y_1 = 1) &= I(z'_{1i}\beta + \alpha y_{2i} + u_i > 0) \\ &= I(\underbrace{z'_{1i}\beta + \alpha y_{2i} + \theta v_i}_{\text{the mean part}} + e_i > 0) \end{aligned}$$

- We will need $D(e|z_1, y_2, v)$.

- First we have

$$\begin{pmatrix} e \\ v \end{pmatrix} = \begin{pmatrix} u - \theta v \\ v \end{pmatrix} = \begin{pmatrix} 1 & -\theta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} \\ \sim N(0, \begin{pmatrix} 1 - \rho^2 & 0 \\ 0 & \sigma_v^2 \end{pmatrix}), \text{ given } z,$$

where $\rho = \text{corr}(u, v) = \sigma_{uv} / \sigma_v$.

- Then e is independent of v : uncorrelatedness of e and v (which also follows from the linear projection), implies independence, since they are jointly normal.

- Let $D(\cdot|\cdot)$ mean conditional distribution

$$\begin{aligned} D(e|z_1, y_2, v) &= D(e|z, v) \text{ (since } y_2 \text{ is generated by } z \text{ and } v) \\ &= D(e|z) \text{ (since } e \text{ is independent of } v) \\ &\sim N(0, 1 - \rho^2). \end{aligned}$$

- So standard assumptions hold: conditional normality, homoskedasticity
- But the identification condition fails: $\text{Var}(e) \neq 1$.
- The outcome equation becomes (CF)

$$y_{1i}^* = z_{1i}'\beta + \alpha y_{2i} + \theta v_i + e_i$$

- Then

$$\begin{aligned}
 P(y_1 = 1|x, v) &= P(y_1 = 1|z_1, y_2, v) \\
 &= P(z'_{1i}\beta + \alpha y_{2i} + \theta v_i + e_i > 0|z_1, y_2, v) \\
 &= P(e_i > -(z'_{1i}\beta + \alpha y_{2i} + \theta v_i)|z_1, y_2, v) \\
 &= \Phi((z'_{1i}\beta + \alpha y_{2i} + \theta v_i)/\sqrt{1-\rho^2})
 \end{aligned} \tag{1}$$

- v_i can be estimated from RF, as $\hat{v}_i$
- A probit of y_1 on z_1, y_2 and $\hat{v}$ gives the estimates, called $\tilde{\beta}$, $\tilde{\alpha}$ and $\tilde{\theta}$.
- So

$$\begin{aligned}
 \tilde{\beta} &\rightarrow \bar{\beta} = \beta/\sqrt{1-\rho^2} \\
 \tilde{\alpha} &\rightarrow \bar{\alpha} = \alpha/\sqrt{1-\rho^2} \\
 \tilde{\theta} &\rightarrow \bar{\theta} = \theta/\sqrt{1-\rho^2}
 \end{aligned}$$

- Recall we are interested in β and α (especially α).
- The problem is now how to estimate ρ .
- Note that

$$\bar{\theta} = \theta / \sqrt{1 - \rho^2}$$

$$\theta = \sigma_{uv} / \sigma_v^2 = \rho \sigma_v / \sigma_v^2 = \rho / \sigma_v$$

- Solve the system for ρ and θ , since σ_v and $\bar{\theta}$ can be estimated.
- So $\sqrt{1 - \rho^2}$ can be estimated: $\sqrt{1 - \rho^2} = \sqrt{1 + \bar{\theta}^2 \sigma_v^2}$.
- So

$$\hat{\beta} = \tilde{\beta} / \sqrt{1 + \tilde{\theta}^2 \hat{\sigma}_v^2}$$

$$\hat{\alpha} = \tilde{\alpha} / \sqrt{1 + \tilde{\theta}^2 \hat{\sigma}_v^2}$$

- An alternative derivation of (1)
- By Fact 1 (see below), $u_i|v_i \sim N(\theta v_i, 1 - \rho^2)$.
- This conditional normality is essentially all we need (we don't really need joint normality)
- So

$$\begin{aligned}
 E(y_{1i}|x_i, v_i) &= P(u_i > -x_i'\beta_0|x_i, v_i) \\
 &= 1 - P(u_i < -x_i'\beta_0|x_i, v_i) \\
 &= 1 - P\left(\frac{u_i - \theta v_i}{\sqrt{1 - \rho^2}} < \frac{-x_i'\beta_0 - \theta v_i}{\sqrt{1 - \rho^2}}|x_i, v_i\right) \\
 &= 1 - \Phi\left(\frac{-x_i'\beta_0 - \theta v_i}{\sqrt{1 - \rho^2}}\right) \\
 &= \Phi\left(\frac{x_i'\beta_0 + \theta v_i}{\sqrt{1 - \rho^2}}\right)
 \end{aligned}$$

Fact 1

- This result is also crucial in Kalman filter.
- If

$$\begin{pmatrix} u \\ v \end{pmatrix} \sim N \left(\begin{pmatrix} m_u \\ m_v \end{pmatrix}, \begin{pmatrix} S_{uu} & S_{uv} \\ S_{vu} & S_{vv} \end{pmatrix} \right)$$

then

$$u|v \sim N(m_{u|v}, S_{u|v})$$

where

$$m_{u|v} = m_u + S_{uv} S_{vv}^{-1} (v - m_v)$$

$$S_{u|v} = S_{uu} - S_{uv} S_{vv}^{-1} S_{vu}$$

- Since v_i has to be estimated, so CF regression involves a *generated regressor*.
- So getting the standard error is hard.
- An alternative approach: MLE (it is also called IV probit)

- Control function approach: in the linear model, it leads to the same estimate as 2SLS
- Wooldridge, Section 6.2, p. 127 & Problem 5.1
- Exercise: show $CF=2SLS$ (using OLS algebra)

When the endogenous variable is continuous

IV probit approach

- IV probit: the key is to obtain $f(y_1, y_2|z)$
- $f(y_1, y_2|z) = f(y_1|y_2, z) \cdot f(y_2|z)$
- First, $f(y_2|z) = f(z_i'\delta + v_i|z) \sim N(z_i'\delta, \sigma_v^2) = \phi((y_2 - z_i'\delta)/\sigma_v)/\sigma_v$
- Second, to get $f(y_1|y_2, z)$, essentially $P(y_1 = 1|y_2, z)$, we need the distribution of u given y_2, z , where $y_{1i}^* = z_{1i}'\beta + \alpha y_{2i} + u_i$

- By Fact 1,

$$f(u|y_2, z) = f(u|z, v) \sim N(\underline{\sigma_{uv}}\sigma_v^{-2}v, 1 - \underline{\sigma_{uv}^2}\sigma_v^{-2})$$

given z

- This can be compared with the exogenous probit, in which case $f(u|y_2, z) = f(u|z, v) \stackrel{u \perp v}{=} f(u|z) \sim N(0, 1)$
- So endogeneity not only introduces non-zero mean but also non-unit variance in the conditional distribution of u .

- So $P(y_1 = 1|y_2, z) =$

$$\begin{aligned}
 &= P(z'_{1i}\beta + \alpha y_{2i} + u_i > 0) = P(u_i > -(z'_{1i}\beta + \alpha y_{2i})) \\
 &= P(u_i - \sigma_{uv}\sigma_v^{-2}v / \sqrt{1 - \rho^2} \\
 &> (-z'_{1i}\beta - \alpha y_{2i} - \sigma_{uv}\sigma_v^{-2}v) / \sqrt{1 - \rho^2}) \\
 &= \Phi((z'_{1i}\beta + \alpha y_{2i} + \rho\sigma_v^{-1}v) / \sqrt{1 - \rho^2}) \\
 &= \Phi((z'_{1i}\beta + \alpha y_{2i} + \underbrace{\rho\sigma_v^{-1}(y_2 - z'\delta)}_{=\theta}) / \sqrt{1 - \rho^2}) \\
 &\equiv \Phi(w)
 \end{aligned}$$

- The likelihood for the i – th observation:

$$l_i = \Phi(w)^{y_1} [1 - \Phi(w)]^{1-y_1} \varphi((y_2 - z'_i\delta)/\sigma_v) / \sigma_v$$

- Log likelihood for the sample $L(\beta, \alpha, \rho, \sigma_v, \delta) = \sum_{i=0}^n \log l_i$
- Maximization over $\beta, \alpha, \rho, \sigma_v, \delta$.
- Standard error: the estimated Hessian, or the outer product of the score.

- A simple test of exogeneity: $\rho = 0$.
- Control function approach: only focuses on $f(y_1|y_2, z)$, is thus a *limited information* procedure.
- CF approach replaces v by $\hat{v}$.

When the endogenous variable is binary

- Binary exogenous variable needs special treatment because the linear model in the first stage is not reasonable any more.
- The model

$$\begin{aligned}y_{1i} &= I\{y_{1i}^* > 0\} \\ y_{1i}^* &= x_i'\beta_0 + u_i = z_{1i}'\beta + y_{2i}'\alpha + u_i \\ y_{2i} &= I\{z_i'\delta + v_i > 0\}\end{aligned}$$

- Like before, we assume (u, v) is jointly normal given z
- But here $\text{Var}(v) = 1$ for identification
- Assume, given z

$$\begin{pmatrix} u_i \\ v_i \end{pmatrix} | z_i \sim N\left(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$$

- The likelihood of observation i : $l_i = f(y_{1i}, y_{2i} | z_i) = f(y_{1i} | y_{2i}, z_i) \cdot f(y_{2i} | z_i)$.
- The second factor is easy to obtain:

$$\begin{aligned} f(y_2 | z) &= P(y_2 = 1)^{y_2} [1 - P(y_2 = 1)]^{1-y_2} \\ &= \Phi(z' \delta)^{y_2} [1 - \Phi(z' \delta)]^{1-y_2} \end{aligned}$$

- The first factor is one of four cases:

$$\begin{aligned} P(y_1 &= 1 | y_2 = 1, z) \\ P(y_1 &= 0 | y_2 = 1, z) [=1 - P(y_1 = 1 | y_2 = 1, z)] \\ P(y_1 &= 1 | y_2 = 0, z) \\ P(y_1 &= 0 | y_2 = 0, z) [=1 - P(y_1 = 1 | y_2 = 0, z)] \end{aligned}$$

- Thus MLE is obtained: $L(\beta, \alpha, \rho, \delta) = \sum_{i=0}^n \log l_i$
- We will calculate these four cases now.

Fact 2

- If v is standard normal,

$$PDF(v|v > a) \sim \varphi(v)/P(v > a) = \varphi(v)/\Phi(-a)$$

- And $PDF(v|v < a) \sim \varphi(v)/P(v < a) = \varphi(v)/[1 - \Phi(-a)]$ (proof?)

When the endogenous variable is binary



$$\begin{aligned} P(y_1 = 1 | y_2 = 1, z) &= E(I(y_1 = 1) | y_2 = 1, z) \\ &= E\{E[I(y_1 = 1) | v, z] | y_2 = 1, z\} \text{ (smaller conditional set dominates)} \\ &= E(P(y_1 = 1 | v, z) | y_2 = 1, z) \\ &\stackrel{*}{=} E(\Phi((z'_1 \beta + \alpha y_{2i} + \rho v) / \sqrt{1 - \rho^2}) | y_2 = 1, z) \\ &= E(\Phi((z'_1 \beta + \alpha y_{2i} + \rho v) / \sqrt{1 - \rho^2}) | v_i > -z'_i \delta, z) \\ &= \int_{\text{support of } v} \Phi((z'_1 \beta + \alpha y_{2i} + \rho v) / \sqrt{1 - \rho^2}) f(v | v_i > -z'_i \delta, z) dv \\ &\stackrel{\text{Fact 2}}{=} \int_{-z'_i \delta}^{\infty} \Phi((z'_1 \beta + \alpha y_{2i} + \rho v) / \sqrt{1 - \rho^2}) \varphi(v) / \Phi(z'_i \delta) dv, \end{aligned}$$

where (*) uses a projection.

- Similarly

$$\begin{aligned}
 P(y_1 = 1 | y_2 = 0, z) &= \int_{-\infty}^{-z'_i \delta} \Phi((z'_{1i} \beta + \alpha y_{2i} + \rho v) / \sqrt{1 - \rho^2}) \varphi(v) / [1 - \Phi(z'_i \delta)] dv
 \end{aligned}$$

- Similarly we can compute $P(y_1 = 0 | y_2 = 1, z)$ and $P(y_1 = 0 | y_2 = 0, z)$

EXAMPLE: Effects of Children on Labor Force Participation, LABSUP.txt

- $y_1 = \text{worked}$, $y_2 = \text{morekids}$ (a dummy for having more than two children).
Population is women with at least two children.

$$\text{worked} = 1[\alpha \cdot \text{morekids} + \beta_0 + \beta_1 \text{nonmomi} + \beta_2 \text{educ} \\ + \beta_3 \text{age} + \beta_4 \text{age}^2 + \beta_5 \text{black} + \beta_6 \text{hispan} + u > 0]$$

- The binary variable *same-sex* is the IV for *morekids*.

Binary panel data model

Pooled probit/logit

- Panel model *without* unobserved effects:

$$P(y_{it} = 1|x_{it}) = \Phi(x'_{it}\beta)$$

- Strict exogeneity (SE): $D(y_{it}|x_i) = D(y_{it}|x_{it})$ for $t = 1, \dots, T$, where $x_i = (x_{i1}, \dots, x_{iT})'$. (D means distribution)
- Stronger than the one in the linear model: in terms of *expectation* there.
- One difficulty of using MLE (which is necessary for binary models) is y_{it} is not independent over t , which is inconvenient in constructing the likelihood function
- Conditional independence (CI): $y_{i1}, y_{i2}, \dots, y_{iT}$ are independent conditional on x_i .

- Under these two assumptions:

$$f(y_{i1}, y_{i2}, \dots, y_{iT} | x_i) \stackrel{CI}{=} \prod_{t=1}^T f(y_{it} | x_i) \stackrel{SE}{=} \prod_{t=1}^T f(y_{it} | x_{it}),$$

where $f(y_{it} | x_{it}) = \Phi(x_{it}\beta)^{y_{it}} [1 - \Phi(x_{it}\beta)]^{1-y_{it}}$.

- Pooled likelihood

$$\sum_{i=1}^n \sum_{t=1}^T [y_{it} \log \Phi(x'_{it}\beta) + (1 - y_{it}) \log(1 - \Phi(x'_{it}\beta))]$$

- Maximizer: pooled probit estimator.
- *Partial likelihood* theory says it is still consistent if CI doesn't hold.

Binary panel data model

RE model

- Now we add unobserved effects
- Response probability:

$$P(y_{it} = 1 | x_{it}, c_i) = \Phi(x'_{it}\beta + c_i)$$

- SE: $D(y_{it} | x_i, c_i) = D(y_{it} | x_{it}, c_i)$ for $t = 1, \dots, T$.
- CI: $y_{i1}, y_{i2}, \dots, y_{iT}$ are independent conditional on x_i, c_i
- $y_{i1}, y_{i2}, \dots, y_{iT}$ are unconditionally dependent because of c_i .

- CI implies:

$$f(y_{i1}, y_{i2}, \dots, y_{iT} | x_i, c_i) \stackrel{CI}{=} \prod_{t=1}^T f(y_{it} | x_i, c_i) \stackrel{SE}{=} \prod_{t=1}^T f(y_{it} | x_{it}, c_i),$$

where $f(y_{it} | x_{it}, c_i) = \Phi(x_{it}\beta + c_i)^{y_{it}} [1 - \Phi(x_{it}\beta + c_i)]^{1-y_{it}}$.

- But c_i is unobservable, so c_i shouldn't appear in the likelihood function
- Viewing c_i as parameters along with β leads to an *incidental parameters* problem.
- It means that MLEs $\hat{c}_i$ and $\hat{\beta}$ are inconsistent. (unlike in the linear case, in which we have $\sqrt{n}$ -consistency)

- Hallmark of FE analysis (in nonlinear models): no specification of a distribution for c_i given x_i .
- RE (random effects) probit: assuming

$$c_i|x_i \sim N(0, \sigma_c^2)$$

(The RE assumption)

- Since the first element of x_i is 1, it implies $c_i \sim N(0, \sigma_c^2)$.
- Then a conditional ML can be applied to β and σ_c^2 as follows.

- Dealing with c_i : integrate out c_i

$$f(y_{i1}, y_{i2}, \dots, y_{iT} | x_i) = \int_{-\infty}^{\infty} \prod_{t=1}^T f(y_{it} | x_{it}, c_i) \phi(c / \sigma_c) / \sigma_c dc.$$

- The likelihood function for the entire sample is then straightforward: production over i
- It is called *RE probit estimator*.
- The estimate is also called *population averaged*.

- As shown before, under RE assumption, viewing $c_i + u_{it}$ as the composite error in the latent variable model and applying pooled probit will give the *attenuation bias*
- This is in contrast to the linear model.
- Latent variable model

$$\begin{aligned} y_i^* &= x'_{it}\beta + c_i + u_{it} \equiv x'_{it}\beta + v_{it} \\ y_{it} &= I(y_i^* > 0) \end{aligned}$$

- $Var(v_{it}|x_{it}) = Var(c_i + u_{it}|x_{it}) \stackrel{u_{it} \perp c_i}{=} Var(u_{it}|x_{it}) + \sigma_c^2 \stackrel{u_{it} \perp c_i}{=} Var(u_{it}|x_{it}, c_i) + \sigma_c^2 = 1 + \sigma_c^2$
- Response probability:

$$P(y_{it} = 1|x_{it}) = \Phi(x'_{it}\beta / \sqrt{1 + \sigma_c^2})$$

- So $\hat{\beta} \xrightarrow{p} \beta / \sqrt{1 + \sigma_c^2}$
- This result also motivates the necessity to specify the distribution of c (in contrast to linear RE model)

Binary panel data model

CRE model

- RE and CI might be too strong.
- Correlated RE (CRE) probit (Chamberlain's): A relaxation of RE:

$$c_i | x_i \sim N(\bar{x}_i' \xi, \sigma_c^2)$$

allowing dependence between c_i and x_i

- Estimation is straightforward: only needs a modification of density of c
-

$$f(y_{i1}, y_{i2}, \dots, y_{iT} | x_i) = \int_{-\infty}^{\infty} \prod_{t=1}^T f(y_{it} | x_{it}, c_i) \varphi((c - \bar{x}_i' \xi) / \sigma_c) / \sigma_c dc.$$

Binary panel data model

FE logit

- Panel logit model:

$$P(y_{it} = 1 | x_{it}, c_i) = G(x'_{it}\beta + c_i),$$

where $G(x) = e^x / (1 + e^x)$.

- An important advantage of panel logit is that we can obtain $\sqrt{N}$ -consistent and asymp. normal estimator of β without any assumptions on $D(c_i | x_i)$, or *achieving fixed effects (FE) estimation*.
- Assumptions: SE, CI, and that each element of x_{it} is time-varying.
- Define $N_i = \sum_{t=1}^T y_{it}$.
- It turns out that $D(y_i | x_i, c_i, N_i)$ does not depend on c_i .
- The functional form of logit helps.
- We can't do this in probit.

- Consider $T = 2$. Then $N_i \in \{0, 1, 2\}$.
- Note that

$$P(y_{i1} = 1 | x_i, c_i, N_i = 0) = 0$$

$$P(y_{i2} = 1 | x_i, c_i, N_i = 0) = 0$$

$$P(y_{i1} = 1 | x_i, c_i, N_i = 2) = 1$$

$$P(y_{i2} = 1 | x_i, c_i, N_i = 2) = 1$$

neither of which is informative for β .

- So we only consider $N_i = 1$.
- As we will show, consistent estimation of β can be obtained by a *standard logit* of y_{i2} on $x_{i2} - x_{i1}$ using the observations for which $N_i = 1$.

$$P(y_{i2} = 1 | x_i, c_i, N_i = 1) = \frac{P(y_{i2} = 1, N_i = 1 | x_i, c_i)}{P(N_i = 1 | x_i, c_i)}$$

where

$$\text{Top} = P(y_{i2} = 1, y_{i1} = 0 | x_i, c_i) \stackrel{CI}{=} P(y_{i2} = 1 | x_i, c_i) P(y_{i1} = 0 | x_i, c_i)$$

$$\stackrel{SE}{=} \frac{G(x'_{i2}\beta + c_i)[1 - G(x'_{i1}\beta + c_i)]}{(1 + e^{x'_{i2}\beta + c_i})(1 + e^{x'_{i1}\beta + c_i})},$$

$$\text{Bottom} = P(y_{i2} = 1, y_{i1} = 0 | x_i, c_i) + P(y_{i2} = 0, y_{i1} = 1 | x_i, c_i)$$

$$= G(x'_{i2}\beta + c_i)[1 - G(x'_{i1}\beta + c_i)] + [1 - G(x'_{i2}\beta + c_i)]G(x'_{i1}\beta + c_i)$$

$$= \frac{e^{x'_{i1}\beta + c_i} + e^{x'_{i2}\beta + c_i}}{(1 + e^{x'_{i2}\beta + c_i})(1 + e^{x'_{i1}\beta + c_i})}.$$

- So

$$\begin{aligned}
 P(y_{i2} = 1 | x_i, c_i, N_i = 1) &= \frac{e^{x'_{i2}\beta + c_i}}{e^{x'_{i1}\beta + c_i} + e^{x'_{i2}\beta + c_i}} = \frac{e^{(x_{i2} - x_{i1})'\beta}}{1 + e^{(x_{i2} - x_{i1})'\beta}} \\
 &= G((x_{i2} - x_{i1})'\beta)
 \end{aligned}$$

- c_i is canceled!
- On the other hand

$$\begin{aligned}
 P(y_{i1} = 1 | x_i, c_i, N_i = 1) &= P(y_{i2} = 0 | x_i, c_i, N_i = 1) \\
 &= 1 - G((x_{i2} - x_{i1})'\beta)
 \end{aligned}$$

$$f(y_{i1}, y_{i2} | x_i, c_i, N_i = 1)$$

$$\begin{aligned} &= P(y_{i1} = 1 | x_i, c_i, N_i = 1) \text{ or } P(y_{i2} = 1 | x_i, c_i, N_i = 1) \\ &= P(y_{i1} = 1 | x_i, c_i, N_i = 1)^{y_{i1}} P(y_{i2} = 1 | x_i, c_i, N_i = 1)^{y_{i2}} \end{aligned}$$

- So the conditional log likelihood function for observation i is

$$\begin{aligned} l_i(\beta) &= I(N_i = 1) [y_{i1} \log(1 - G((x_{i2} - x_{i1})' \beta)) \\ &\quad + y_{i2} \log G((x_{i2} - x_{i1})' \beta)] \\ y_{i1} = 1 - y_{i2} \quad I(N_i = 1) &= [(1 - y_{i2}) \log(1 - G((x_{i2} - x_{i1})' \beta)) \\ &\quad + y_{i2} \log G((x_{i2} - x_{i1})' \beta)] \end{aligned}$$

- We select out the observations for which $N_i = 1$.
- Computationally, it is just a *standard logit* of y_{i2} on $x_{i2} - x_{i1}$ using the observations for which $N_i = 1$.

- It is called conditional MLE (CMLE), as it is different for a full MLE.
- To get a full MLE: $P(y_{i1}, y_{i2}|x_i, c_i)$

$$\begin{aligned}
 &= \underbrace{P(y_{i1}, y_{i2}|x_i, c_i, N_i = 0)}_{\text{deterministic}} P(N_i = 0|x_i, c_i) \\
 &+ \underbrace{P(y_{i1}, y_{i2}|x_i, c_i, N_i = 1)}_{\text{CMLE}} P(N_i = 1|x_i, c_i) \\
 &+ \underbrace{P(y_{i1}, y_{i2}|x_i, c_i, N_i = 2)}_{\text{deterministic}} P(N_i = 2|x_i, c_i).
 \end{aligned}$$

- Here we used the rule of total probability:
 $P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2)$ if B_1 and B_2 forms a partition of the sample space.
- A side: this rule was used to show the selection bias, which mistakenly assumes $P(A) = P(A|B_1)$.
- CMLE doesn't consider $P(N_i = 0|x_i, c_i)$, $P(N_i = 1|x_i, c_i)$ or $P(N_i = 2|x_i, c_i)$ which may depend on β .